FIITJEE Students From Classroom / Integrated School Programs have secured to 2 Zonal, 6 State & 18 City Topper Ranks. 33 in Top 100, 78 in Top 200 and 205 in Top 500 All India Ranks bagged by FIITJEE Students from All Programs have qualified in JEE (Advanced), 2015.

ALL INDIA TEST SERIES

FIITJEE

JEE (Advanced)-2016

FULL TEST – I

Paper 1

Time Allotted: 3 Hours **Maximum Marks: 246**

- Please read the instructions carefully. You are allotted 5 minutes specifically for this purpose.
- You are not allowed to leave the Examination Hall before the end of the test.

INSTRUCTIONS

A. General Instructions

1. Attempt ALL the questions. Answers have to be marked on the OMR sheets.
2. This question paper contains Three Parts.
3. **Part-I** is Physics, **Part-II** is Chemistry and **Part-III** is Mathematics.
4. Each part is further divided into three sections: **Section-A, Section-B & Section-C**
5. Rough spaces are provided for rough work inside the question paper. No additional sheets will be provided for rough work.
6. Blank Papers, clip boards, log tables, slide rule, calculator, cellular phones, pagers and electronic devices, in any form, are not allowed.

B. Filling of OMR Sheet

1. Ensure matching of OMR sheet with the Question paper before you start marking your answers on OMR sheet.
2. On the OMR sheet, darken the appropriate bubble with black pen for each character of your Enrolment No. and write your Name, Test Centre and other details at the designated places.
3. OMR sheet contains alphabets, numerals & special characters for marking answers.

C. Marking Scheme For All Three Parts.

1. **Section – A (01 – 07)** contains 7 multiple choice questions which have only one correct answer. Each question carries **+3 marks** for correct answer and **– 1 mark** for wrong answer.

   Section – A (08 – 11) contains 4 multiple choice questions which have one or more than one correct answer. Each question carries **+3 marks** for correct answer. There is no negative marking.

   Section-A (12 – 16) contains 2 paragraphs. Based upon paragraph, 2 and 3 multiple choice questions have to be answered. Each question has only one correct answer and carries **+3 marks** for correct answer and **– 1 mark** for wrong answer.

2. **Section – B (01 – 02)** contains 2 Matrix Match Type questions containing statements given in 2 columns. Statements in the first column have to be matched with statements in the second column. Each question carries **+8 marks** for all correct answer. For each correct row **+2 marks** will be awarded. There may be one or more than one correct matching. No marks will be given for any wrong matching in any question. There is no negative marking.

3. **Section-C (01 – 06)** contains 6 Numerical based questions with answers as numerical value from **0 to 9** and each question carries **+3 marks** for correct answer. There is no negative marking.

Name of the Candidate	
Enrolment No.	

Useful Data

PHYSICS

Acceleration due to gravity	$g = 10\ m/s^2$
Planck constant	$h = 6.6 \times 10^{-34}$ J-s
Charge of electron	$e = 1.6 \times 10^{-19}$ C
Mass of electron	$m_e = 9.1 \times 10^{-31}$ kg
Permittivity of free space	$\varepsilon_0 = 8.85 \times 10^{-12}\ C^2/N\text{-}m^2$
Density of water	$\rho_{water} = 10^3\ kg/m^3$
Atmospheric pressure	$P_a = 10^5\ N/m^2$
Gas constant	$R = 8.314\ J\ K^{-1}\ mol^{-1}$

CHEMISTRY

Gas Constant	R	=	$8.314\ J\ K^{-1}\ mol^{-1}$
		=	0.0821 Lit atm $K^{-1}\ mol^{-1}$
		=	$1.987 \approx 2$ Cal $K^{-1}\ mol^{-1}$
Avogadro's Number	N_a	=	6.023×10^{23}
Planck's constant	h	=	6.625×10^{-34} J·s
		=	6.625×10^{-27} erg·s
1 Faraday		=	96500 coulomb
1 calorie		=	4.2 joule
1 amu		=	1.66×10^{-27} kg
1 eV		=	1.6×10^{-19} J

Atomic No: H=1, He = 2, Li=3, Be=4, B=5, C=6, N=7, O=8, N=9, Na=11, Mg=12, Si=14, Al=13, P=15, S=16, Cl=17, Ar=18, K =19, Ca=20, Cr=24, Mn=25, Fe=26, Co=27, Ni=28, Cu = 29, Zn=30, As=33, Br=35, Ag=47, Sn=50, I=53, Xe=54, Ba=56, Pb=82, U=92.

Atomic masses: H=1, He=4, Li=7, Be=9, B=11, C=12, N=14, O=16, F=19, Na=23, Mg=24, Al = 27, Si=28, P=31, S=32, Cl=35.5, K=39, Ca=40, Cr=52, Mn=55, Fe=56, Co=59, Ni=58.7, Cu=63.5, Zn=65.4, As=75, Br=80, Ag=108, Sn=118.7, I=127, Xe=131, Ba=137, Pb=207, U=238.

Physics

PART – I

SECTION – A

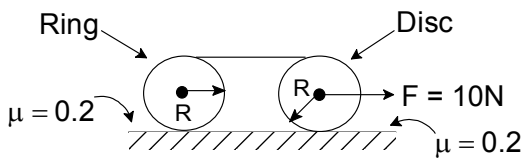
Straight Objective Type

This section contains **7 multiple choice** questions. Each question has four choices (A), (B), (C) and (D) out of which **ONLY ONE** option **is correct**.

1. A particle strikes a horizontal frictionless floor with a speed u at an angle θ with the vertical and rebounds with a speed v at an angle ϕ with the vertical. The co-efficient of restitution between the particle and floor is e. The angle ϕ is equal to

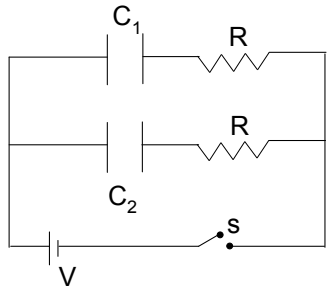
(A) θ (B) $\tan^{-1}(e\tan\theta)$ (C) $\tan^{-1}\left(\frac{1}{e}\tan\theta\right)$ (D) $(1+e)\theta$

2. In the given situation disc and ring (2kg) are connected with a string. Both are placed on the rough surface. A force F = 10N is applied on the centre of disc horizontally. Assume initially both bodies were in rest. Then frictional force acting on the ring will be



(A) $\frac{20}{3}$N
(B) $\frac{10}{3}$N
(C) Zero
(D) None

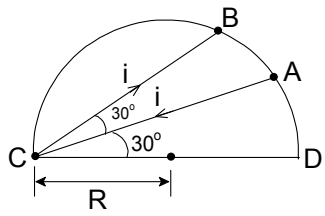
3. In the circuit shown $C_1 = 2C_2$. Switch S is closed at time t = 0. Let i_1 & i_2 be the current flowing through C_1 and C_2 at any time t, then the ratio I_1/I_2
(A) is constant
(B) increases with increase in time t
(C) decreases with increase in time t
(D) first increases then decreases



Space for Rough work

4. The capacitor of an oscillatory circuit of frequency 10^4Hz is enclosed in a container. When the container is evacuated, the frequency changes by 50 Hz, the dielectric constant of the gas is
(A) 1.1 (B) 1.01 (C) 1.001 (D) 1.0001

5. A current carrying wire is placed in the grooves of an insulating semicircular disc of radius 'R' as shown. The current enters at point A and leaves point B. Determine the magnetic field at point D.

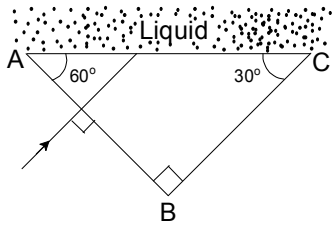


(A) $\frac{\mu_0 i}{8\pi R\sqrt{3}}$ (B) $\frac{\mu_0 i}{4\pi R\sqrt{3}}$
(C) $\frac{\sqrt{3}\mu_0 i}{4\pi R}$ (D) None

6. The organ pipes, both closed at one end, have lengths ℓ and $(\ell + \Delta\ell)$ vibrates in fundamental mode. Neglect end correction. If the velocity of sound in air is v, then the number of beats/sec is $(\text{assume } \Delta\ell << \ell)$

(A) $\frac{v}{4\ell}$ (B) $\frac{v}{2\ell}$ (C) $\frac{v\Delta\ell}{4\ell^2}$ (D) $\frac{v\Delta\ell}{2\ell^2}$

7. Light is incident normally on face AB of a prism as shown. A liquid of refractive index μ is placed on face AC of the prism. The prism is made of glass of refractive index 3/2. The limit of μ for which total internal reflection takes place on face AC is



(A) $\mu > \frac{3}{4}$ (B) $\mu < \frac{3\sqrt{3}}{4}$
(C) $\mu > \sqrt{3}$ (D) $\mu < \frac{\sqrt{3}}{2}$

Space for Rough work

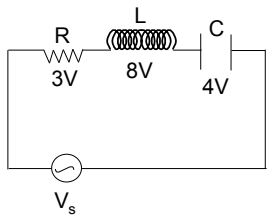
(Multiple Correct Choice Type)

This section contains **4 multiple choice** questions. Each question has four choices (A), (B), (C) and (D) out of which **ONE or MORE THAN ONE** is/are correct.

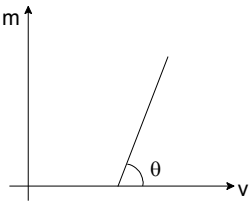
8. A photon collides with a stationary H – atom in ground state inelastically. Energy of the colliding photon is 10.2 eV. Almost instantaneously, another photon collides with same H – atom inelastically with an energy of 15 eV. What will be observed by the detector?
(A) two photons of energy 10.2 eV
(B) two photons of energy 1.4 eV
(C) One electron having kinetic energy nearly 11.6 eV
(D) One photon of energy 10.2 eV and an electron of energy 1.4 eV

9. An electron in H – atom is excited from ground state level to first excited level. Select the correct statements:
(A) Kinetic energy of electron is decreased
(B) Potential energy is increased while kinetic energy of electron is decreased
(C) Time period of rotation and angular momentum both are increased
(D) Total energy is increased while angular speed is decreased

10. In the given circuit, current in resistance R is 1A, then
(A) $V_s = 5$volt
(B) impedance of network is 5Ω
(C) Power factor of given circuit is 0.6 lagging
(D) Power factor of given circuit is 0.6 leading



11. Figure shows variation of magnification m(produced by a thin convex lens) and distance v of image from pole of the lens. Which of the following statements are correct?
(A) focal length of the lens is equal to intercept on v - axis
(B) focal length of the lens is equal to inverse of slope of the line
(C) Magnitude of intercept on m – axis is equal to unity
(D) None



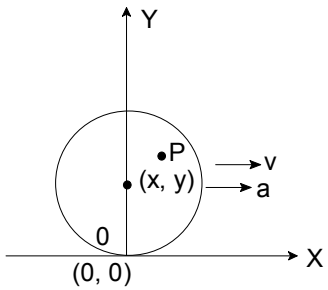
Space for Rough work

Comprehension Type

This section contains **2 paragraphs.** Based upon one of the paragraphs **2 multiple choice questions** and based on the other paragraph **3 multiple choice question** have to be answered. Each of these questions has four choices (A), (B), (C) and (D) out of which **ONLY ONE** is correct.

Paragraph for Questions 12 & 13

A disc having radius R is rolling without slipping on a horizontal plane as shown. Centre of the disc has a velocity v and acceleration a as shown.



12. Speed of point P having coordinates (x, y) is

(A) $\frac{V\sqrt{x^2+y^2}}{R}$ (B) $\frac{V\sqrt{x^2+(y+R)^2}}{R}$ (C) $\frac{V\sqrt{x^2+(y-R)^2}}{R}$ (D) None

13. If $V=\sqrt{2aR,}$ the angle θ between acceleration of the topmost point and the horizontal is

(A) 0 (B) 45° (C) $\tan^{-1}(2)$ (D) $\tan^{-1}\left(\frac{1}{2}\right)$

Paragraph for Questions 14 & 16

A certain substance has a mass of 50g/mol. When 300J of heat is added to 25 gm of sample of this material, its temperature rises from 25 to 45^oC.

14. The thermal heat capacity of the substance is

(A) $25J/^oC$ (B) $5J/^oC$ (C) $15J/^oC$ (D) $20J/^oC$

15. The specific heat capacity of the substance is

(A) $300J/kg^oC$ (B) $400J/Kg^oC$ (C) $500J/Kg^oC$ (D) $600J/kg^oC$

16. The molar heat capacity of the substance is

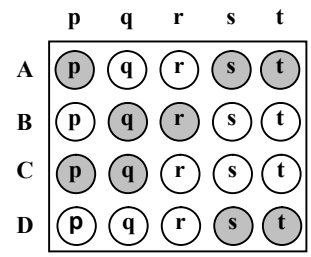
(A) $30J/mol^oC$ (B) $40J/mol^oC$ (C) $20J/mol^oC$ (D) $60J/mol^oC$

Space for Rough work

SECTION - B
Matrix – Match Type

*This section contains 2 questions. Each question contains statements given in two columns, which have to be matched. The statements in **Column I** are labelled A, B, C and D, while the statements in **Column II** are labelled p, q, r, s and t. Any given statement in **Column I** can have correct matching with **ONE OR MORE** statement(s) in **Column II.** The appropriate bubbles corresponding to the answers to these questions have to be darkened as illustrated in the following example:*

If the correct matches are A – p, s and t; B – q and r; C – p and q; and D – s and t; then the correct darkening of bubbles will look like the following:



1. Match the following

Column – I		Column – II	
(A)	Zero gravity	(p)	SHM
(B)	$\sqrt{\frac{k}{m}}$ of a spring	(q)	$Y = \pm\frac{A}{\sqrt{2}}$
(C)	KE = PE	(r)	$\frac{1}{3}$rd the time period of $y = \sin\omega t$
(D)	$y = \sin(3\omega t)$	(s)	writst watch can give correct time
		(t)	Per second

2. Match the following

Column – I		Column – II	
(A)	Nuclear fusion	(p)	Stratifies $E = mc^2$
(B)	Nuclear fission	(q)	Generally possible for nuclei with low atomic number
(C)	β – decay	(r)	Generally possible for nuclei with higher atomic number and instability
(D)	Exothermic nuclear reaction	(s)	Essentially proceeds by weak nuclear force
		(t)	Significant momentum conservation

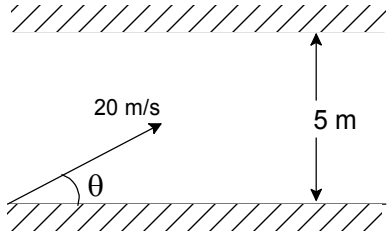
Space for Rough work

SECTION – C

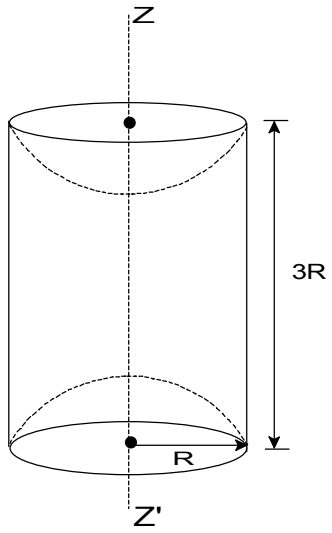
(Integer Answer Type)

This section contains **6 questions**. Each question, when worked out will result in **one integer** from 0 to 9 (both inclusive).

1. A projectile is projected with a speed 20 m/s from the floor of a 5 m high room as shown. The maximum horizontal range of the projectile is found to be $20\sqrt{n}$ m. Find n [Take g = 10 ms^{-2}].

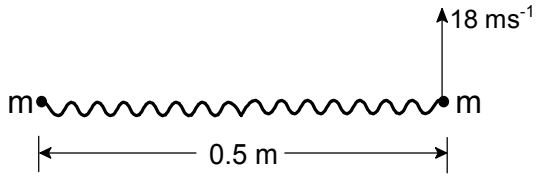


2. A solid cylinder is made of radius R and height 3R having mass density ρ. Now two half spheres of radius R are removed from both ends. The moment of intertia of remaining portion about axis ZZ' can be calculated as $\frac{29}{6K}\pi R^5\rho$. Find K.



Space for Rough work

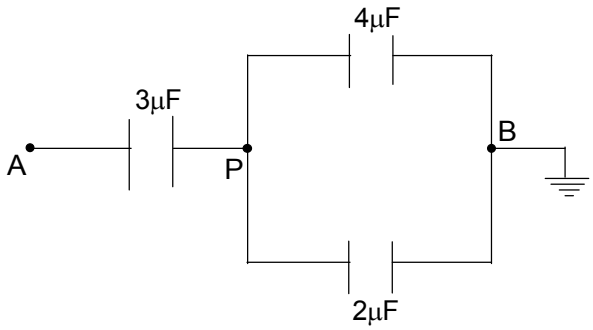
3. Two particles each of mass m are connected by a string of length 1m. The particles are kept on a smooth horizontal plane. The initial separation between the particles is 0.5 m. One of the particles is given a velocity 18 m/s as shown. Find the magnitude of angular velocity of one particle with respect to the other particle just after the string becomes taut (in m/s).



4. Two thin symmetrical lenses of different nature have equal radii of curvature of all faces R = 20 cm. The lenses are put close together and immersed in water. The focal length of the system is 24 cm. The difference between refractive indices of the lenses is $\left(\frac{5}{n}\right)$. Find n. The refractive index of water is $\frac{4}{3}$.

5. A 10 m long nichrome wire having 80Ω resistance has current carrying capacity of 5 A. This wire can be cut into equal parts and equal parts can be connected in series or parallel. What is the maximum power which can be obtained as heat by the wire from a 200 V mains supply (in KW).

6. In the given figure, a potential of +12 volt is given to point A, and point B is earthed. What is the potential at point P in volt?



Space for Rough work

Chemistry

PART – II

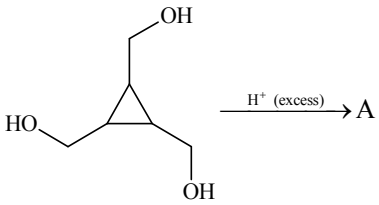
SECTION – A

Straight Objective Type

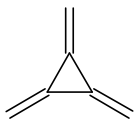
This section contains **7 multiple choice** questions. Each question has 4 choices (A), (B), (C) and (D) for its answer, out which **ONLY ONE** option is correct.

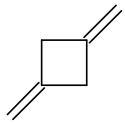
1. Which one of the following is not true about diborane?
   (A) It has two bridging hydrogens and four terminal hydrogens.
   (B) When methylated, the product is $Me_4B_2H_2$.
   (C) The bridging hydrogen are in a plane perpendicular to the rest of the B – H bonds.
   (D) All the six B – H bond distances are equal.

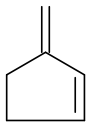
2. For the reaction,

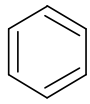


What is the product A?

(A) 

(B) 

(C) 

(D) 

Space for Rough work

3. The wave function of atomic orbital of H-like atoms is given as under:

$$\psi_{2s} = \frac{1}{4\sqrt{2}\pi} Z^{3/2} (2 - Zr) e^{-\frac{Zr}{2}}$$

Given that the radius is in Å, then which of the following is radius for nodal surface for He^+ ion

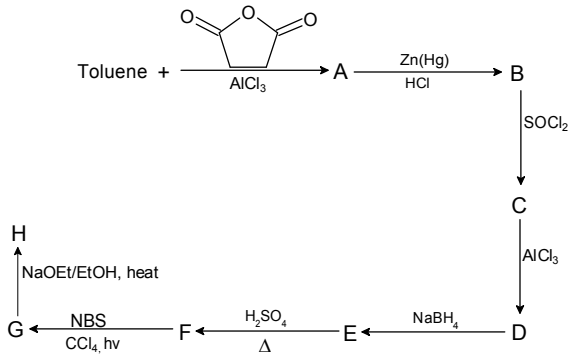
(A) 1Å (B) 2Å (C) 2.5Å (D) 4Å

4. Cyclopentadiene + Na $\longrightarrow$ C + H_2

$2C + FeCl_2 \longrightarrow D + 2NaCl$

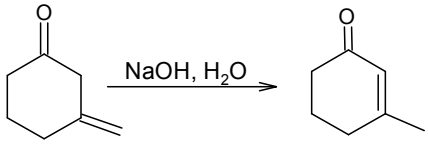
Which one of the following in not true about D.
(A) D is an organometalic compound
(B) D is an optically inactive compound
(C) EAN of Fe^{+2} in D is 34
(D) All C – C bonds in D are of identical length

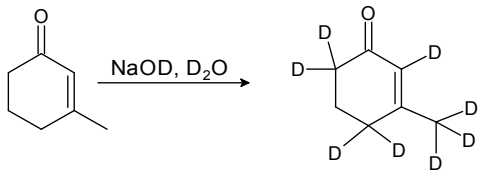
5. Which one of following statement is true
(A) G is an optically inactive compound
(B) H is an optically inactive compound and have all C – C bond of equal length
(C) F and D both have equal no of π-bond
(D) H is 3-methylnaphthalene

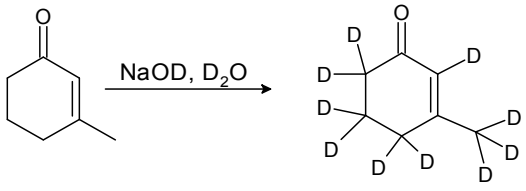


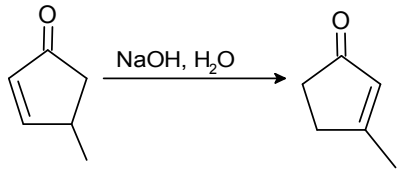
Space for Rough work

6. Which one of the following reaction is not possible:-

(A) 

(B) 

(C) 

(D) 

7. (I) $Co(CO)_6$
(II) $Co(CO)_5NH_3$
(III) $Co(CO)_4(NH_3)_2$
(IV) $Co(CO)_3(NH_3)_3$
(V) $Co(CO)_2(NH_3)_4$
(VI) $Co(CO)(NH_3)_5$
Which statement is correct for these compounds?
(A) order of Co – C bond length in complexes is I > II > III > IV > V > VI
(B) order of Co – C bond length in complexes is VI > V > IV > III > II > I
(C) order of Co – C bond strength in complexes is I = II = III = IV = V = VI
(D) order of C – O bond length in complexes is I = II = III = IV = V = VI

Multiple Correct Choice Type

This section contains 4 multiple choice questions. Each question has 4 choices (A), (B), (C) and (D) for its answer, out which **ONE OR MORE** is/are correct.

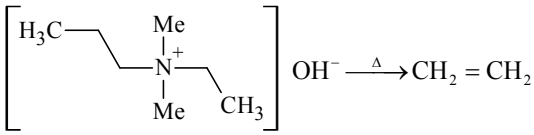
8. Which of the following is/are correct for the reaction with equilibrium constant K?

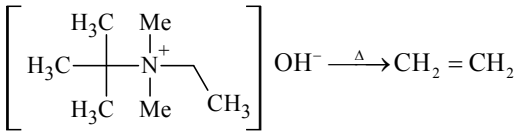
$$A(g)+B(g)\underset{K_b}{\overset{K_f}{\rightleftharpoons}}C(g)+D(g);\quad \Delta H^{\circ}=20J$$

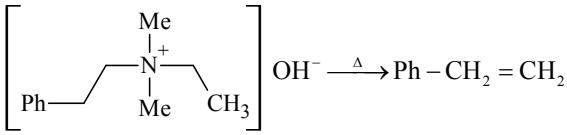
(A) K_{eq} will increase on increasing temperature
(B) K_f will increase on increasing temperature
(C) K_b will increase on increasing temperature
(D) K_b will decrease on increasing temperature

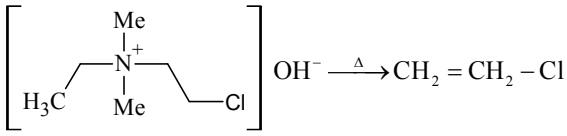
Space for Rough work

9. Which of the following reaction is incorrect regarding the formation of major product (alkene)?

(A) 

(B) 

(C) 

(D) 

10. Which among the following statements is/are correct?
(A) pH of 10^{-8} M HCl is equal to 8
(B) Conjugate base of $H_2PO_4^-$ is HPO_4^{2-}
(C) pH of 0.1 M NaCl (aqueous solution) = $\frac{1}{2}pK_w$
(D) Ionization of water increases with decrease in temperature

11. 4.5 g of Al (atomic mass = 27) is deposited at cathode from Al^{+3} solution by certain quantity of charge. The correct statement/s is/are:
(A) The volume of H_2 produced at 0°C and 1 atm pressure from H^+ ions in solution by the same quantity of charge will be 5.6 ℓ.
(B) The volume of H_2 produced at 0°C and 1 atm pressure from H^+ ions in solution by the same quantity of charge will be 22.4 ℓ.
(C) The number of moles of Al deposited will be 0.166 moles
(D) The number of faraday used for reduction of 1 mole of Al^{+3} is $\frac{3}{18}$F.

Space for Rough work

Comprehension Type

This section contains **2 paragraphs.** Based upon the first paragraph 2 multiple choice questions and based upon the second paragraph **3 multiple choice questions** have to be answered. Each of these questions has four choices A), B), C) and D) out of **WHICH ONLY ONE IS CORRECT.**

Paragraph for Question Nos. 12 to 13

On the basis of aromaticity, there are three types of compounds i.e. aromatic, non-aromatic and anti-aromatic. The increasing order of stability of these compounds are is under:

Anti-aromatic compound $<$ non-aromatic compound $<$ aromatic compounds. Compounds to be aromatic follow the following conditions (according to valence bond theory)

(i) The compounds must be be cyclic in structure having $(4n + 2)\pi\ e^-$, where n = Hückel's number = 0, 1, 2, 3 et.c

(ii) The each atoms of the cyclic structure must have unhybridised p-orbital i.e. the atoms of the compounds have unhybridised p-orbital i.e. usually have sp^2 hybrid or planar.

(iii) There must be a ring current of π electrons in the ring or cyclic structure i.e. cyclic structure must undergo resonance .

Compounds to be anti-aromatic, it must have $4n\pi e^-$ where n = 1, 2… and it must be planar and undergo resonance. Non-aromatic compounds the name itself spells that compounds must be non-planar irrespective of number of π electrons. Either it has $4n\pi e^-$ or $(4n + 2)\ \pi$ electrons it does not matter.

The rate of reaction of any aromatic compounds depends upon the following factors:

(i) Electron density

(ii) stability of carbocation produced

Higher the amount of electron density of the ring, higher will be its rate towards aromatic electrophilic substitution and vice-versa. Similarly, higher will be the stability of the produced carbocation after the attack of electrophile, higher will be its rate towards aromatic electrophilic substitution. There is a great effect of kinetic labelling on the rate of aromatic electrophilic substitution. As we known that higher the atomic weight or, molecualr weight, higher will be the van der Waal's force of attraction or, bond energy. Since there will be effect of kienetic labelling if the 2nd step of the reaction will be the slow step, (r.d.s.) otherwise there will be no effect of kinetic labelling.

12. Which of the following is correct order of the rate of reaction of C_6H_6, C_6D_6 and C_6T_6 towards sulphonation?

(A) Same rate of reactions of C_6H_6, C_6D_6 and C_6T_6

(B) $C_6T_6 > C_6D_6 > C_6H_6$

(C) $C_6H_6 > C_6D_6 > C_6T_6$

(D) $C_6H_6 > C_6D_6 = C_6T_6$

Space for Rough work

13. Which of the following is the correct order of rate of reaction of C_6H_6, C_6D_6 and C_6T_6 towards nitrations?
(A) $C_6H_6 > C_6D_6 > C_6T_6$ (B) $C_6H_6 = C_6D_6 = C_6T_6$
(C) $C_6H_6 > C_6D_6 = C_6T_6$ (D) $C_6T_6 > C_6D_6 > C_6H_6$

Paragraph for Question Nos. 14 to 16

The Bertholet's equation of state for 1 mole real gas is given as under:

$$\left(P + \frac{a}{TV}\right)(V - b) = RT$$

Where a and b are van der Waal's constants. van der Waal's constant "a" signifies, the force of attraction among the gas molecules. van der Waal's constant "b" signifies incompressible volume i.e. volume having no effect of compression and expansion. It is also known as co-volume. Thereby the volume having effect of compression and expansion i.e. compressible volume = V – b (for 1 mole real gas).

One of the form of van der Waal's equation of state is virial equation. The virial equation for 1 mole real gas is as under:

$$PV = RT\left[1 + \frac{A}{V} + \frac{B}{V^2} + \frac{C}{V^3} + \ldots \text{to higher power of n}\right]$$

where A, B, C are known as 2nd, 3rd and 4th virial coefficients respectively. The temperature at which real gases obey ideal gas equation (PV = RT), is known as Boyles temperature i.e. T_B.

14. If $T_i = \frac{2a}{Rb}$, where T_i is the inversion temperature, then which of the following is the correct relation of the Boyle's temperature T_B obtained by the Bertholet's equation

(A) $T_i = 2T_B^2$ (B) $T_i = 2T_B$ (C) $2T_i = T_B$ (D) $T_i^2 = T_B$

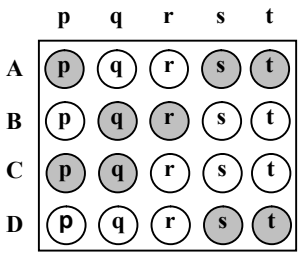
Space for Rough work

15. If the value of a and b increases by 8 times and 2 times respecively, then new Boyle's temperature will increases by how many times and what will be effect on the ease of liquification ?
(A) 2 times, ease of liquification will increase
(B) 4 times, the ease of liquification will increase
(C) $\frac{1}{4}$ times, the ease of liquification will decrease
(D) $\frac{1}{2}$ times, the ease of liquification will decrease

16. If the third virial coefficient of 4 moles of mono-atomic as is 0.04 $(lit/mol)^2$ at NTP, then what will be the real gas at NTP?
(A) 88.8 liters (B) 80 liters (C) 89 liters (D) 90.4 liters

SECTION-B
(Matrix-Match Type)

This section contains 2 questions. Each question contains statements given in two columns, which have to be matched. The statements in **Column I** are labelled A, B, C and D, while the statements in **Column II** are labelled p, q, r, s and t. Any given statement in **Column I** can have correct matching with **ONE OR MORE** statement(s) in **Column II.** The appropriate bubbles corresponding to the answers to these questions have to be darkened as illustrated in the following example:
If the correct matches are A – p, s and t; B – q and r; C – p and q; and D – s and t; then the correct darkening of bubbles will look like the following:



1. Match the following.

Column – I (Carbides)		Column – II (Hydrolysis of compound in Column – I)	
(A)	SiC	(p)	Resulting gas does not produce red ppt with ammonical Cu_2Cl_2
(B)	Al_4C_3	(q)	Unsaturated hydrocarbon is produced
(C)	CaC_2	(r)	No hydrocarbon produced
(D)	Mg_2C_3	(s)	Resulting gas produce white ppt. with Tollen's reagent
		(t)	Staturated hydrocarbon is produced

Space for Rough work

2. Match the solution given in column I with the properties given in column II.

Column – I		Column - II	
(A)	0.1 M $Al_2(SO_4)_3$	(p)	Solution with highest being point
(B)	0.1 M $AlPO_4$	(q)	Van't Hoff factor in greater than 1
(C)	0.1 M Urea	(r)	Solution with lowest osmotic pressure
(D)	0.1 M $MgCl_2$	(s)	Solution with lowest freezing point
		(t)	Solution with lowest boiling point

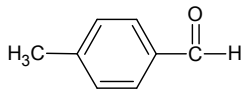
SECTION –C
Integer Answer Type

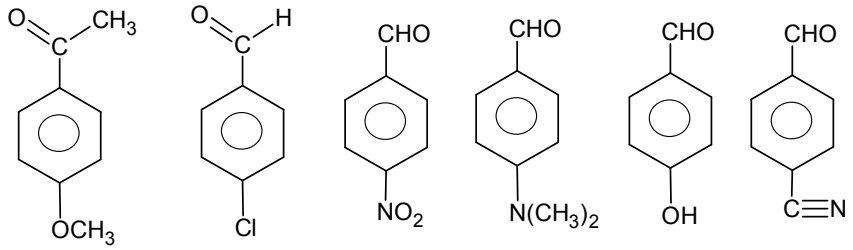
This section contains **6 questions**. Each question, when worked out will result in **one integer** from 0 to 9 (both inclusive).

1. The number of bicarbonates that do not exist in solid form among the following is.....
$LiHCO_3, NaHCO_3, Ca(HCO_3)_2, KHCO_3, NH_4HCO_3, Ba(HCO_3)_2, Mg(HCO_3)_2$.

2. In Borax
(i) Number of B – O – B bond is x
(ii) Number of B – B bond is y
(iii) Number of sp^2 hybridized B atom is z
Calculate the value of x + y + z

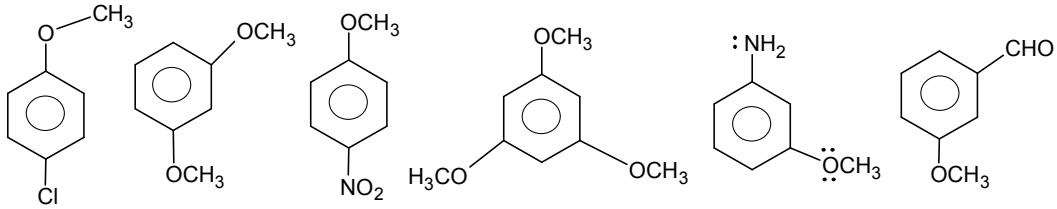
3. Examine the structural formulas of following compounds and find out number of compounds which show
higher rate of nucleophilic addition than





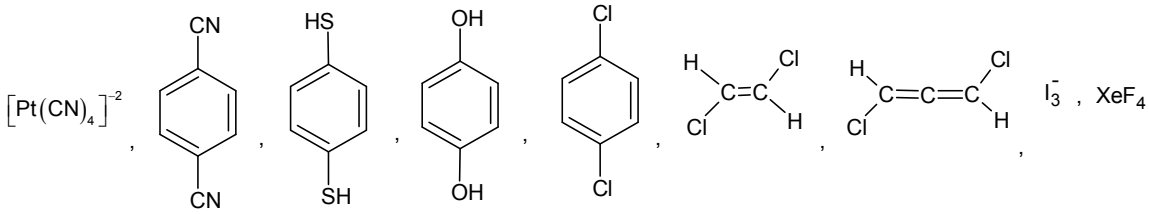
Space for Rough work

4. Examine the structural fromuals shown below and identify how many compounds will show coupling reaction with diazonium salt faster than anisole



5. For the first order reaction $t_{99\%} = x \times t_{90\%}$
The value of x will be

6. Total no of molecules/ions which have non zero dipole moment is



Space for Rough work

Mathematics

PART – III

SECTION – A

Straight Objective Type

This section contains 7 multiple choice questions numbered 1 to 7. Each question has 4 choices (A), (B), (C) and (D), out of which **ONLY ONE** is correct

1. If 'P' be a point on unit circle with centre at point 'O'. If Q is another point inclined at angle $\frac{\pi}{3}$ to 'OP' and $\left|\overrightarrow{OT}\right| = \left|\overrightarrow{OQ}\right| = r$, then the interval in which 'r' lies, so that $2\left|\overrightarrow{AT}\right| = \left|\overrightarrow{PQ}\right|$, and $\overrightarrow{OA} = \hat{a}$, is,

(A) $\left[\frac{7+\sqrt{13}}{6}, \infty\right)$ (B) $\left[\frac{7-\sqrt{13}}{6}, \frac{7+\sqrt{13}}{6}\right]$

(C) $\left(-\infty, \frac{7-\sqrt{13}}{6}\right]$ (D) None of these

2. If a bag contains five balls. Two balls are drawn and are found to be white. What is the probability that all the balls are white?

(A) $\frac{1}{4}$ (B) $\frac{1}{3}$ (C) $\frac{1}{2}$ (D) $\frac{2}{3}$

3. If f(x) and g(x) are defined and differentiable for $x \geq x_0$ and $f(x_0) = g(x_0), f'(x) > g'(x)\ \forall x > x_0$, then

(A) $f(x) > g(x)$, for some $x > x_0$ (B) $f(x) > g(x)$, for no $x > x_0$

(C) $f(x) = g(x)$, for some $x > x_0$ (D) $f(x) > g(x)$, for all $x > x_0$

4. If z_1 and z_2 are two variable complex numbers such that, $|z_1|^2 \leq 169$ & $|z_2 + 3 - 4i|^2 \leq 25$. Then the maximum value of $|z_1 - z_2|$,is

(A) 23 (B) 13 (C) 5 (D) 18

Space for Rough work

5. If $S=\frac{4}{1.2.3}+\frac{5}{2.3.4}+\frac{6}{3.4.5}+\frac{7}{4.5.6}$..... up to 'n' terms & $S=K-\frac{(2n+5)}{2(n+1)(n+2)}$, Then,

(A) $K=\frac{1}{4}$ (B) $K=\frac{1}{2}$ (C) $K=\frac{3}{4}$ (D) $K=\frac{5}{4}$

6. The perpendicular from origin to the tangent at any point on a curve is equal to the obscissa of the point of contact. If the curve passes through point (1, 2), then it also passes through point,

(A) $(1,-2)$ (B) $(1,3)$ (C) $(2,3)$ (D) $(-2,1)$

7. For an ellipse $\frac{x^2}{9}+\frac{y^2}{4}=1$, with vertices A & A', tangent drawn at the point 'P' in the first quadrant meets the y – axis in Q and the chord A'P meets the y – axis in M. If 'O' is the centre of ellipse, then $OQ^2 - MQ^2$ is equal to,

(A) 9 (B) 13 (C) 4 (D) 5

Multiple Correct Choice Type

This section contains 4 multiple choice questions. Each question has 4 choices (A), (B), (C) and (D) for its answer, out of which **ONE OR MORE** is/are correct

8. If $f(x)=x^5+x^4+x^3+x^2+x+3$. Then f(x) = 0 has,

(A) odd number of real roots

(B) Atleast one root lies between $(-\infty,-1)$

(C) Greatest integer value of at least one root = 2

(D) All roots are negative

9. Which of the following is/are not true,

(A) if $\{f(x)+g(x)\}$ is continuous at x = a, then f(x) and g(x) are both separately continuous at x = a.

(B) If f(x) .g(x) is continuous at x = a, then f(x) and g(x) are separately continuous at x = a

(C) If f(x). g(x) & f(x) are discontinuous at x = a then g(x) is continuous at x = a

(D) if both f(x) and g(x) are discontinuous at x = a then f(x) + g(x) may not be discontinuous at x = a

Space for Rough work

10. If $\int \tan^{-1}\left(\frac{\cos 2x}{1+\sin 2x}\right)dx = g(x)+c$, then

(A) g(x) represent an equation of parabola, whose concave side is upward

(B) g(x) have point of extrema at $x = \frac{\pi}{4}$

(C) g(x) represent an equation of parabola, whose concave side is downward

(D) Minimum value of g(x) = $\frac{\pi^2}{32}$

11. If α,β are the roots of equation, $x^2-2x-a^2+1=0$ and γ,δ are the roots of the equation, $x^2-2(a+1)x+a(a-1)=0$ such that $\alpha,\beta \in (\gamma,\delta)$, then the interval in which 'a' lies,

(A) $a \in \left(-\frac{1}{4},1\right)$ (B) $a \in \left(-\frac{1}{8},\frac{1}{2}\right)$ (C) $a \in (-\infty,1)$ (D) $a \in (-\infty,-2)$

Comprehension Type

This section contains 2 paragraphs. Based upon one of paragraphs 2 multiple choice questions and based on the other paragraph 3 multiple choice questions have to be answered. Each of these questions has four choices (A), (B), (C) and (D) out of which **ONLY ONE** is correct

Paragraph for Question Nos. 12 to 13

Consider the curves,

$C_1 \equiv y^2 = 4K_1x$, $\forall K_1 \in \left[\frac{1}{8},\frac{1}{4}\right]$

$C_2 \equiv x^2 = 4K_2(y-2)$, $\forall K_2 \in \left[-1,-\frac{1}{4}\right]$

$C_3 \equiv x = 0$

12. Maximum area bounded by all of these three curve is,

(A) $\frac{7}{3}-\sqrt{2}$ (B) $\frac{7}{3}+\sqrt{2}$ (C) 2 (D) 1

13. Minimum area bounded by all of these three curve is,

(A) $\frac{7}{3}-\sqrt{2}$ (B) $\frac{7}{3}+\sqrt{2}$ (C) 2 (D) 1

Space for Rough work

Paragraph for Question Nos. 14 to 16

Triangle ABC has circumcentre 'O' and orthocentre 'H'. Points E and F are chosen on sides AC and AB respectively such that AE = AO and AF = AH. If 'R' be radius of circle circumscribing the triangle ABC. Then,

14. The area of quadrilateral BFEC is equal to,

(A) $\frac{R^2}{4}(\sin B+\sin C)$ (B) $\frac{R^2}{8}[\sin A]\cos(B-C)$

(C) $\frac{R^2}{2}\sin 2A$ (D) $\frac{R^2}{2}(\sin 2B+\sin 2C)$

15. The length of segment EF is equal to,

(A) $R\cos A$ (B) R (C) $R\sin 2A$ (D) $2R\cos A$

16. Let $\angle AFE=\frac{\pi}{3}$, then sum of squares of three altitudes of triangle AFE is equal to,

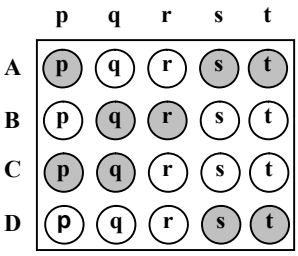
(A) $\frac{9R^2}{4}$ (B) $2R^2$ (C) $\frac{3R^2}{2}$ (D) $\frac{8R^2}{3}$

Space for Rough work

SECTION-B
(Matrix-Match Type)

This section contains 2 questions. Each question contains statements given in two columns, which have to be matched. The statements in **Column I** are labelled A, B, C and D, while the statements in **Column II** are labelled p, q, r, s and t. Any given statement in **Column I** can have correct matching with **ONE OR MORE** statement(s) in **Column II.** The appropriate bubbles corresponding to the answers to these questions have to be darkened as illustrated in the following example:

If the correct matches are A – p, s and t; B – q and r; C – p and q; and D – s and t; then the correct darkening of bubbles will look like the following:



1. Match the following:

Column – I		Column – II	
(A)	Let $\theta_1,\theta_2,\theta_3,\theta_4$ be eccentric angles of four concylic points of ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$, and $\theta_1+\theta_2+\theta_3+\theta_4=K\pi$, then K can be,	(p)	2
(B)	Normals to parabola $y^2=4x$ at P and Q meets at $R(x_2,0)$, if $x_2=4$ and area of circle Circumscribing ΔPQR is $K\pi$ then K is equal to,	(q)	9
(C)	Let $S\equiv x^2+y^2-6x-8y+21\le 0$. Then, max $\left[\left\{\frac{12x}{7}-\frac{5y}{7}\right\};x,y\in s\right]$	(r)	4
(D)	In triangle ABC let $D\left(0,\sqrt{3}\right),E(1,0),F(-1,0)$ be the feet of $\perp^r$ dropped from A, B, C to opposite sides, Then square of the sum of ordinates of point A, B, and C is	(s)	3
		(t)	6

Space for Rough work

2. Match the following:

Column – I		Column – II	
(A)	The solution of 1 + secx = $\cot^2\frac{x}{2}$ Is given by $x = K\pi$, then K is/are	(p)	2
(B)	If α,β are roots of equation $A\left(x^2+m^2\right)+Amx+Cm^2x^2=0$ then $A\left(\alpha^2+\beta^2\right)+A\alpha\beta+C\alpha^2\beta^2+3$ is/are equal to	(q)	3
(C)	The sum of binomial co – efficient in expansion of $\left(x^{-\frac{3}{4}}+nx^{\frac{5}{4}}\right)^m$ where m is positive integer lies between 200 and 400 & the term independent of x equal to 448, then n^5 divisible by	(r)	4
(D)	If last digit of the number 2^{999} is α and last digit of $3^{5\alpha}$ is μ, then $\mu+\alpha=$	(s)	8
		(t)	9

SECTION – C

(Integer Answer Type)

This section contains **6 questions**. Each question, when worked out will result in **one integer** from 0 to 9 (both inclusive).

1. A line makes an angle of $\alpha,\beta,\gamma,\delta$ with the four diagonals of a cube then the value of $\left[\cos^2\alpha+\cos^2\beta+\cos^2\gamma+\cos^2\delta\right]$ is, $\left(\text{where}[.]\text{denotes the GIF}\right)$.....

2. Tangents drawn from any point 'P' on the curve $xy-2x-y-2=0$ intersect the asymptotes at A & B. Then the area of triangle QAB, is 4K (where Q is centre of the curve). Then value of K is……

Space for Rough work

3. Consider a triangle ABC whose sides length are AB = 4, BC = 2 & CA = 3. If the bisector of the angle 'C' cuts AB in D and the circumcircle in E, then the value of $\left[\frac{CE}{DE}\right]$ is……(where, [.] denotes the GIF)

4. If $\int_0^{10} f(x)dx = 5$ & $\sum_{K=1}^{10}\int_0^1 f(K-1+x)dx = R$. Then value of R is……..

5. If the number of ways in which four distinct balls can be put into two identical boxes so that no box remains empty is equal to K, then K =…….

6. The value of $|K|$ for which the following system of equations is consistent,

$(K+1)^3 x + (K+3)^3 y = (K+3)^3$

$(K+1)x + (K+2)y = K+3$

$x + y = 1$

Space for Rough work